Design “Class” Diagram

Full Diagram (Simplified Views)

This diagram shows the connection between all classes: ViewType, Header, the views (ToDo, Calendar, and Habit Tracker), Task, Tag, Filter, and Habit Data. The Header, Task, and views connection will be discussed in the Views Diagram. The Task Class is what most Classes are built around, holding all the values needed for the customization from the user (ID, event, title, tags, description, startDate, endDate, isVisible, and completed). The Tag Class is used by the Task Class as a way to quickly sort through similar tasks, holding an ID, name, color, and completed variables. The Filter Class introduces the ability to search for a specific task you are looking for based on information imputed such as a tag, name, or date. Using the isVisible function, the filter finds tasks that do not match specific criteria and sets their isVisible variable to false to prevent them from displaying on the calendar. The HabitData Class encapsulates analytics for the habit tracker, tracking success rate, tag distribution, tasks done, and unfinished tags. The arrows from View to Filter and HabitData indicate that the View uses these to render filtered and summarized data to the user.

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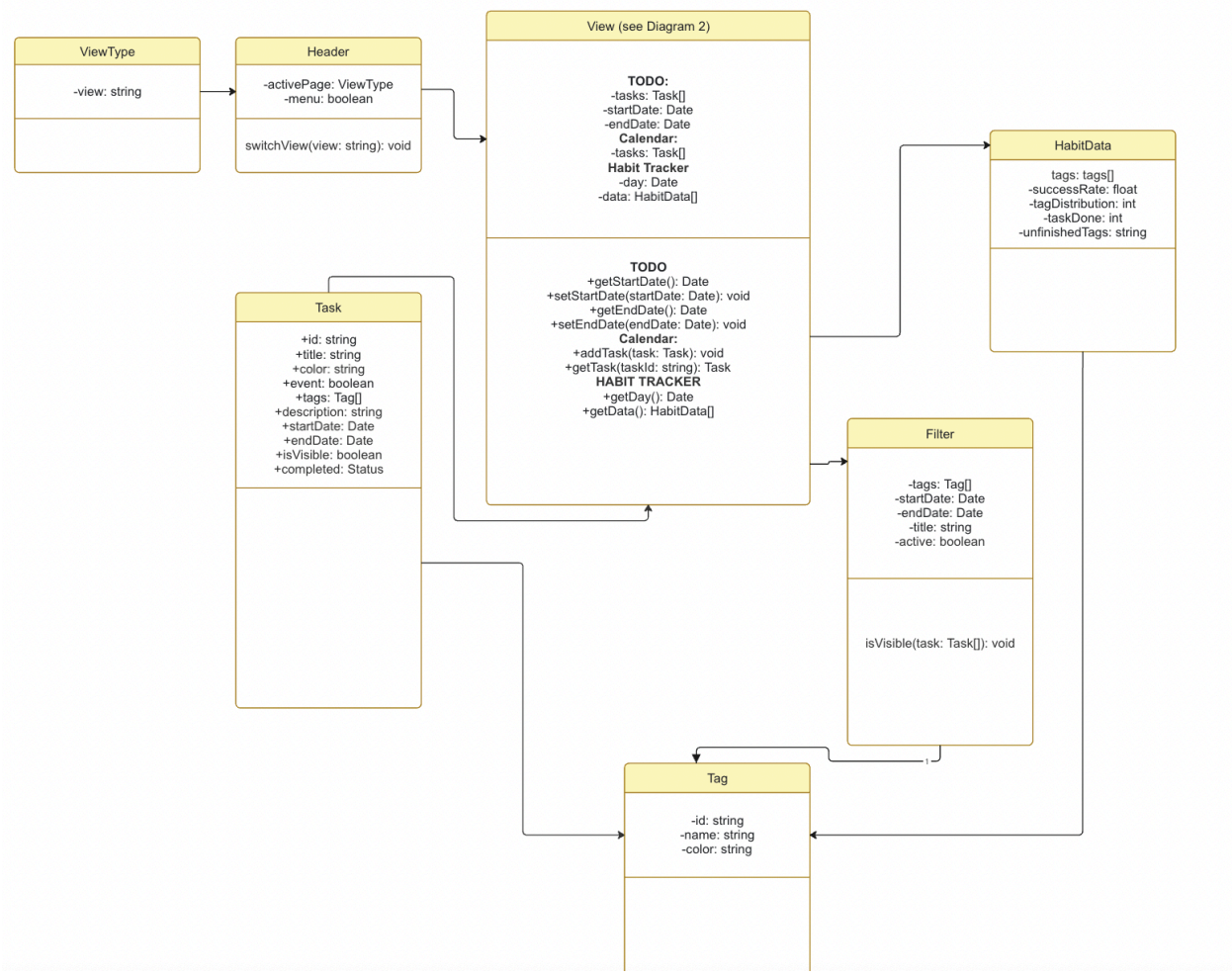

Views Diagram

The diagram shows the main views of the application (Calendar, HabitTracker, and ToDo), with the Header class acting as the router between all of them. The class Header holds the activePage to decide what view it is on and the switchView() command to allow the user to swap between views. The Task Class is what all of the views use to display to the user what they must do on that day, holding all the values needed for the customization from the user (ID, event, title, tags, description, startDate, endDate, isVisible, and completed). The Calendar Class holds a list of Task[] to display all tasks that event is true this is checked using the isEvent function, which sets all non-events' visible value to false. The HabitTracker Class uses the same Task class that is shared amongst the views to display to the user what their habits look like, as seen on Diagram 1 in the HabitData class. The ToDo class displays all tasks with the event variable set to false, the same way Calendar does using isEvent(), in a neat format for a span of days, depending on the startDate and endDate variables.

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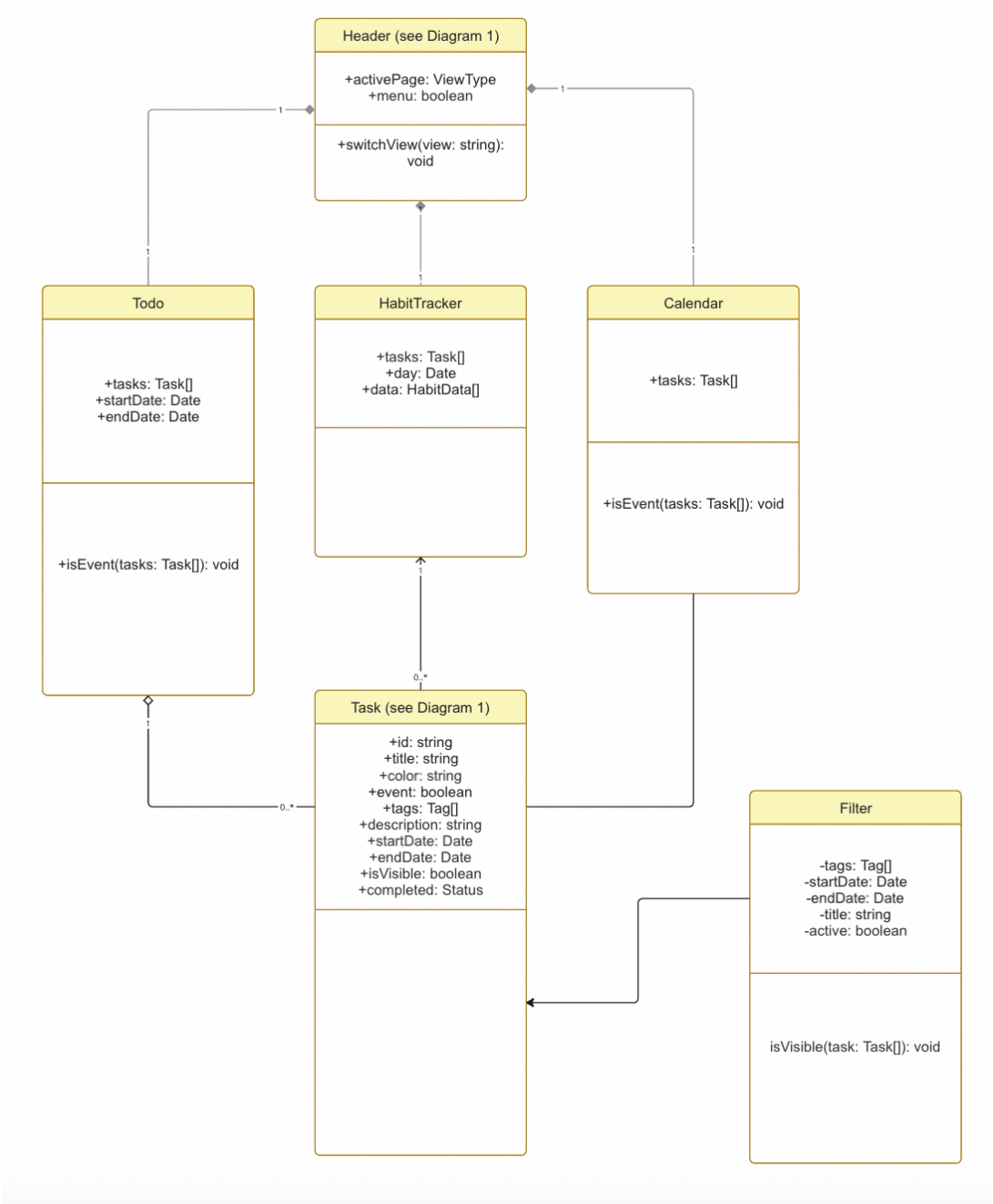
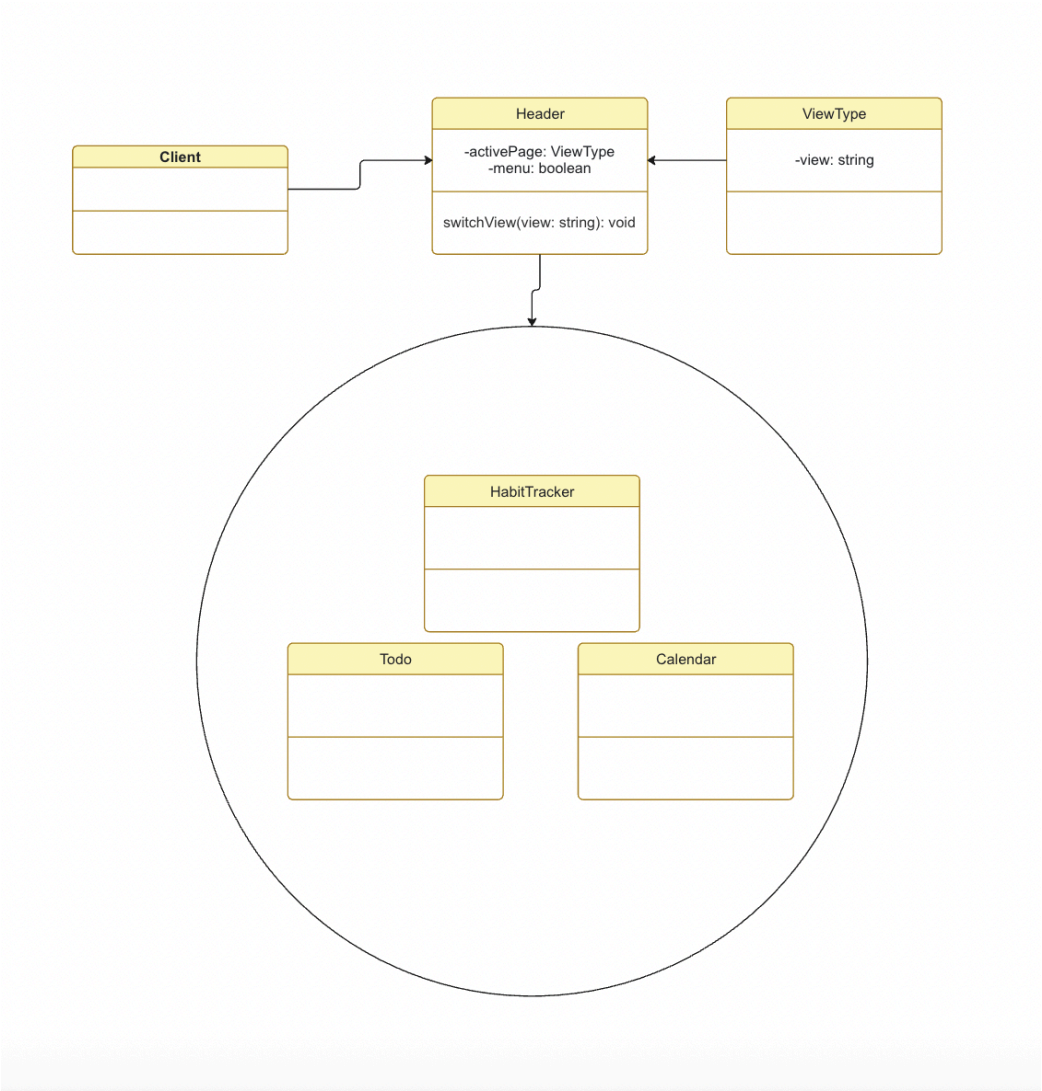


Design Patterns

Facade

The Header class functions as a Facade over the three view subsystems (Todo, HabitTracker, and Calendar). The Header class acts as the sole router (or, in Troy's explanation, the remote control) for the major view components using switchView(view: String). This decision was made as it simplifies the complexity and keeps all routing in a single component class rather than having it all over the place. On top of that, if any future views are added later, you would only have to go to one component to fix the routing on it.

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